Zulham Amin Ristianto

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SUMMARY PROFILE

I have completed my Bachelor's Applied Degree in Automation Engineering Technology at Diponegoro University with a GPA of 3.85/4.00 and am currently awaiting graduation. I have hands-on experience in industrial automation, industrial IoT, and manufacturing digitalization through internships at PT Toyota Motor Manufacturing Indonesia and PT Mesin Isuzu Indonesia. Experienced in developing industrial monitoring systems, data acquisition, IoT dashboards, PLC-based automation, electrical wiring, and engineering documentation. Strong technical background in PLC programming, microcontrollers, Industrial IoT, SCADA, Node-RED, MQTT, and industrial communication protocols. Proven leadership experience through student organizations and project teams, demonstrating strong communication, collaboration, and problem-solving skills. Passionate about industrial automation, smart manufacturing, and continuous improvement, with the ability to adapt quickly, learn new technologies, and contribute effectively in dynamic engineering environments.

EDUCATION

University of Diponegoro 2022 - 2026
A new student at University of Diponegoro. Majoring Automation Engineering Technology.

- **GPA :** 3.85 / 4.0
- **Relevant Coursework :** Microcontroller, Internet Of Things (Iot) PLC, SCADA, Basic Electronics, Pneumatic And Hydraulic.

WORK EXPERIENCES

PT. Toyota Motor Manufacturing Indonesia **Oktober 2025 – January 2026**
Plant 1 Sunter – Engine Production
Internship – Maintenance Division

- Developed a TPM (Total Productive Maintenance) digitalization system to replace manual maintenance records and improve maintenance data tracking.
- Designed and implemented a digital Andon system to support faster response during machine downtime in the engine production line.
- Assisted in electrical schematic reading and electrical drawing for maintenance and automation-related activities.
- Supported maintenance and troubleshooting of production machines under supervision of maintenance engineers.
- Gained hands-on experience in industrial automation systems, maintenance workflows, and Toyota Production System (TPS) environment.

Key Skills: TPM digitalization, Power BI, Power Apps, i-Reporter, Andon system, Industrial maintenance, Electrical drawing, Automation systems, PLC basics, Troubleshooting, Manufacturing automation.

PT. Mesin Isuzu Indonesia **August – September 2025**
Internship – Maintenance Division

- Developed a data acquisition monitoring project for CYB20 and CYB50 machines using Moxa and ABB devices.
- Assisted in electrical wiring and electrical drawing for machine maintenance and improvement projects.
- Designed and implemented a Node-RED based dashboard for real-time monitoring and

visualization of machine data.
- Gained hands-on experience in industrial automation, troubleshooting, and electrical systems.

Key Skills: Electrical wiring, Electrical drawing, Node-RED programming, Dashboard design, Data acquisition systems.

ORGANIZATION EXPERIENCES

Secretary of the Daily Executive Board 2024 – 2025
Student Association of Automation Engineering Technology
Responsible for assisting in the coordination of internal meetings, managing documentation and correspondence, and ensuring the smooth operation of organizational administration.

Head of Organizing Committee – Student Association Roadshow 2023 – 2024
Student Association of Automation Engineering Technology
Led the planning and execution of a campus-wide roadshow event aimed at promoting student organization activities and increasing student participation in academic and non-academic programs.

Assessment Division Staff – AEROSPACE Event 2024
Participated in the assessment team responsible for evaluating participants' performance and feedback in the AEROSPACE (Academic and Engineering Student Conference) event.

PROJECT

Smart Andon System Project – PT. Toyota Motor Manufacturing Indonesia
- Developed a Smart Andon system to improve response time during machine downtime in the engine production line.
- Replaced conventional/manual Andon activation with a digital and automated alert system based on machine status.
- Designed the system logic and alert flow to notify maintenance and related teams more efficiently.
- Assisted in integrating machine signals with the Andon system for real-time condition monitoring.
- Supported maintenance operations by enabling faster issue identification and escalation.

TPM Digitalization Project (i-Reporter) – PT. Toyota Motor Manufacturing Indonesia
- Designed and structured digital TPM checklists to ensure consistency with existing maintenance standards.
- Improved data accuracy, traceability, and accessibility of maintenance records by eliminating manual paper documentation.
- Supported maintenance teams in adopting the digital TPM workflow to increase efficiency and reduce reporting time.
- Assisted in monitoring maintenance activities through real-time digital reporting and documentation.

Data Acquisition System for Real-Time Monitoring of Current, Battery, and Machine Operational Status (IoT-Based) – PT. Mesin Isuzu Indonesia
- Developed an IoT-based data acquisition system for real-time monitoring of current (ampere), battery condition, and operational status of CYB350 machines.
- Integrated Moxa industrial gateway for data communication between field devices and monitoring system.
- Utilized ABB SCU to acquire and process machine electrical and operational data.
- Designed and implemented a Node-RED dashboard for real-time data visualization and monitoring.

Power supply 12V AC-DC Conveter

- Choosing the components to be used to be effective
- Assemble components to match their function
- Soldering components to connect with each other so that power supply can be used.

Design and Development of an ESP32-Based Flood Detection and Mitigation System with GPS Integration, Actuator Control, and Telegram Notifications

- Choosing the components to be used to be effective
- Assemble components to match their function
- Soldering components to connect with each other so that power supply can be used.
- Implementing IoT communication for online data management.

PLC Ladder Diagram Programming for Beverage Can Press Machine Using Omron PLC

- Designed and simulated a ladder diagram program for an automated beverage can pressing system using Omron PLC.

Data Acquisition and Interface System for Distance Detection Using Ultrasonic Sensor and Delphi7 Application

- Created a data acquisition system that measures distance using an ultrasonic sensor, interfaced through a Delphi7-based desktop application**.**

Design and Development of an IoT-Based Smart Lock System Using ESP32, Facial Recognition, and Telegram Notifications for Enhanced Security

Role: Software Developer:

- Developed the software architecture for a smart locking system using the ESP32 microcontroller.
- Integrated face recognition technology for secure access authentication.
- Programmed real-time notification system via Telegram API to alert users of access events.
- Implemented efficient communication between hardware and cloud-based services for remote monitoring and control.

SKILLS, ACTIVITIES & INTERESTS

Languages: Active in Indonesian; Passive in English
Technical Skills: Microsoft Office (Word, Excel, PowerPoint), MATLAB, CX-Programmer, FluidSIM, Proteus, Eagle, PSIM, Dev C++, Arduino IDE, Fusion 360, AutoCAD, Delphi
Soft Skills : Teamwork, Leadership, Problem-solving, Time Management, Adaptability, Critical Thinking, Communication, Analytical Thinking.

CERTIFICATION

- **TOEFL ITP** – Score: **622** 15 Juni 2026
- **Basic Safety Awareness** – Toyota Indonesia Academy 2025
- **Industrial Internship Certificate** – PT Toyota Motor Manufacturing Indonesia 2025
- **Industrial Internship Certificate** – PT Mesin Isuzu Indonesia 2025